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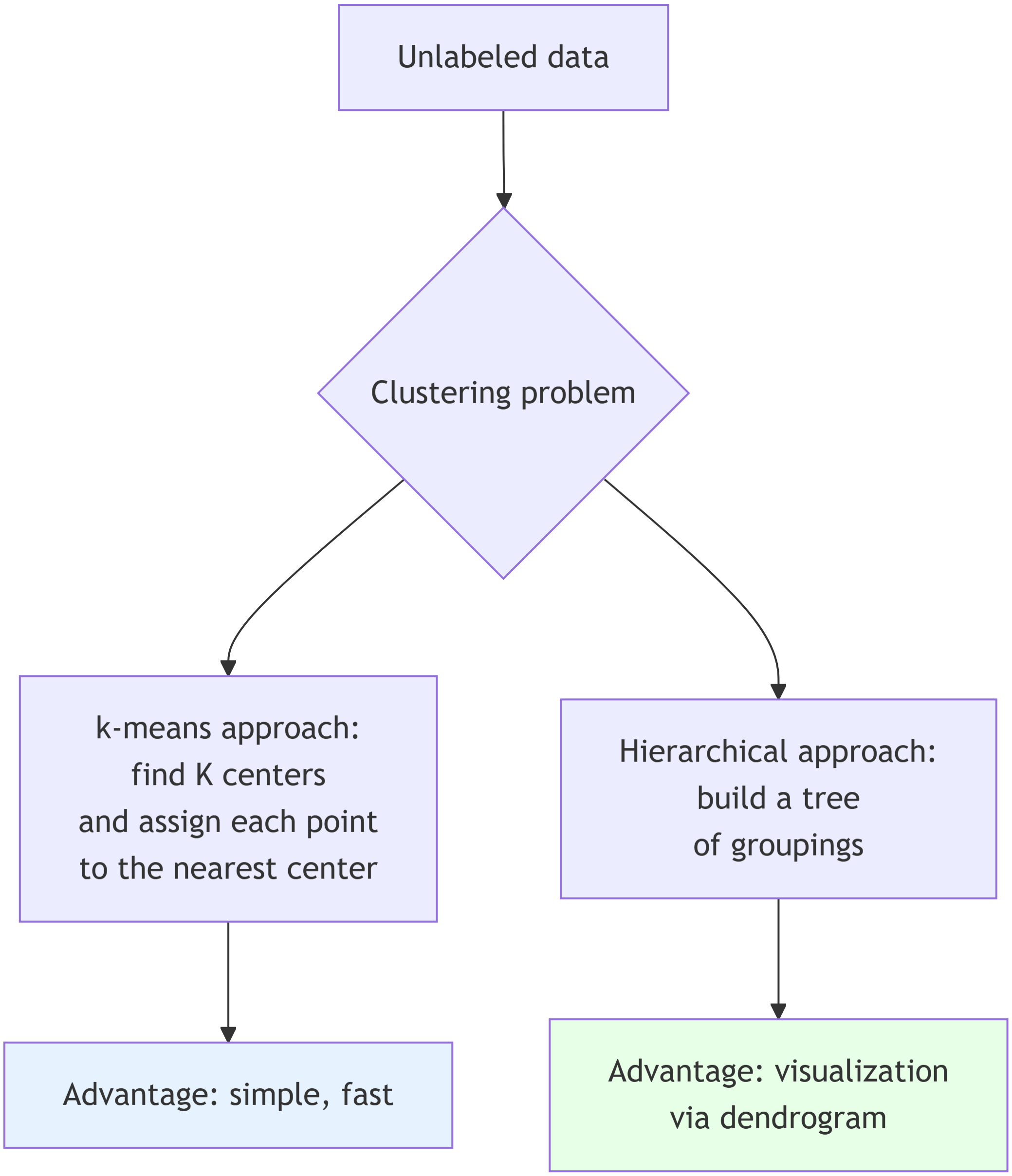
Intuition

1. The Fundamental Problem: Grouping Without a Guide

Imagine this: You're given 1,000 unlabeled animal photos and asked: "How many different types of animals are there, and which photos go together?" This is exactly the clustering problem!

K-means answers this question by making **a simplifying assumption**: each type of animal can be represented by an **average image** (prototype), and each photo is close to one of these prototypes.

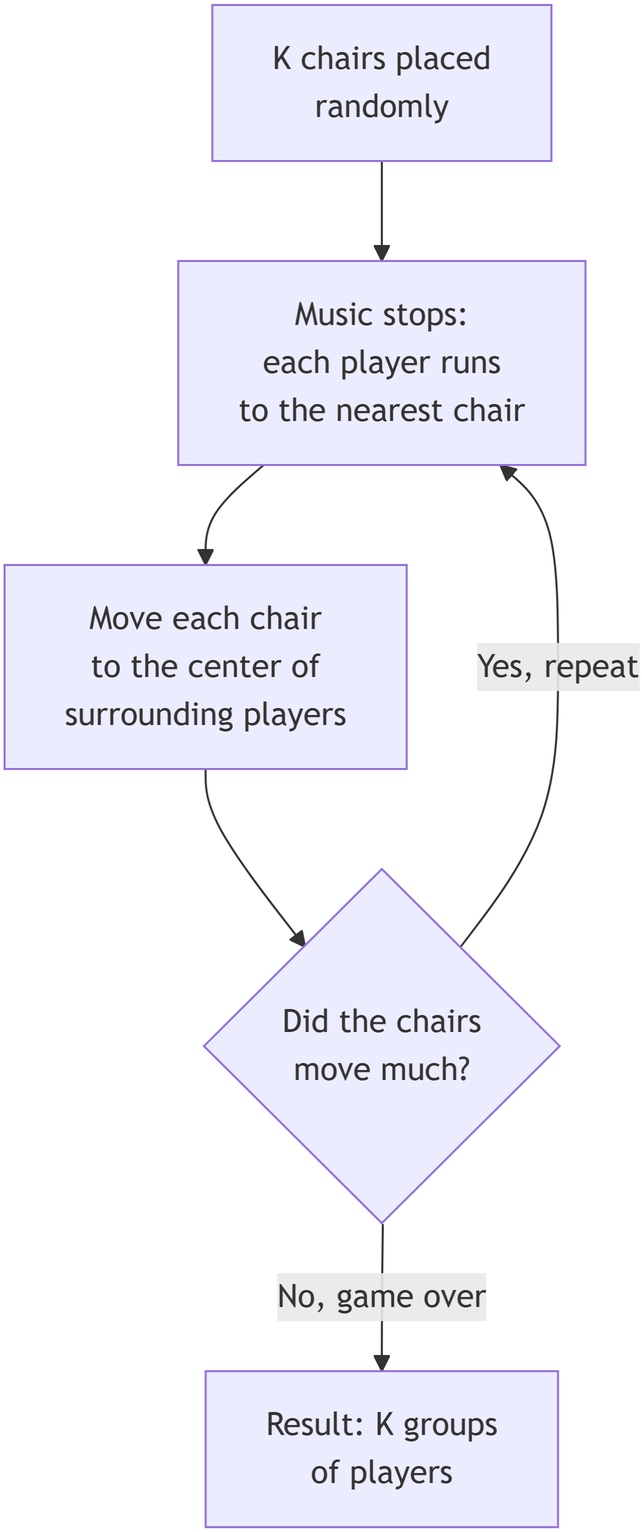
> **!** Important
>
> **Key Concept: Unsupervised Learning** Clustering is **unsupervised**: no labels, no teacher to say "this is a cat." The algorithm must discover the structure by itself.



2. The Geometric Intuition: The Musical Chairs Game

Analogy: Imagine a game of musical chairs with K chairs (clusters) and N players (data points). When the music stops: 1. Each player runs to the nearest chair **(assignment)** 2. Each chair is moved to the center of the players surrounding it **(update)** 3. Repeat until the chairs no longer move

This is exactly k-means!


K chairs placed
randomly
Music stops:
each player runs
to the nearest chair
Move each chair
to the center of
surrounding players
Yes, repeat
Did the chairs
move much?
No, game over
Result: K groups
of players

3. The Mathematical Formulation: Minimizing Energy

The Criterion to Minimize

K-means seeks to minimize:

```
L = sum over all points of (distance to its cluster center)²
```

Why squared? Two reasons: 1. **Mathematical**: makes the problem easier to solve 2. **Geometric**: penalizes more points that are very far away

The Two Magic Equations

Assignment step (E-like): For each point `x_i`, find the nearest center ` _k`:

```
Assign x_i to cluster k that minimizes ||x_i -  _k||²
```

Update step (M-like): For each cluster `k`, recalculate its center:

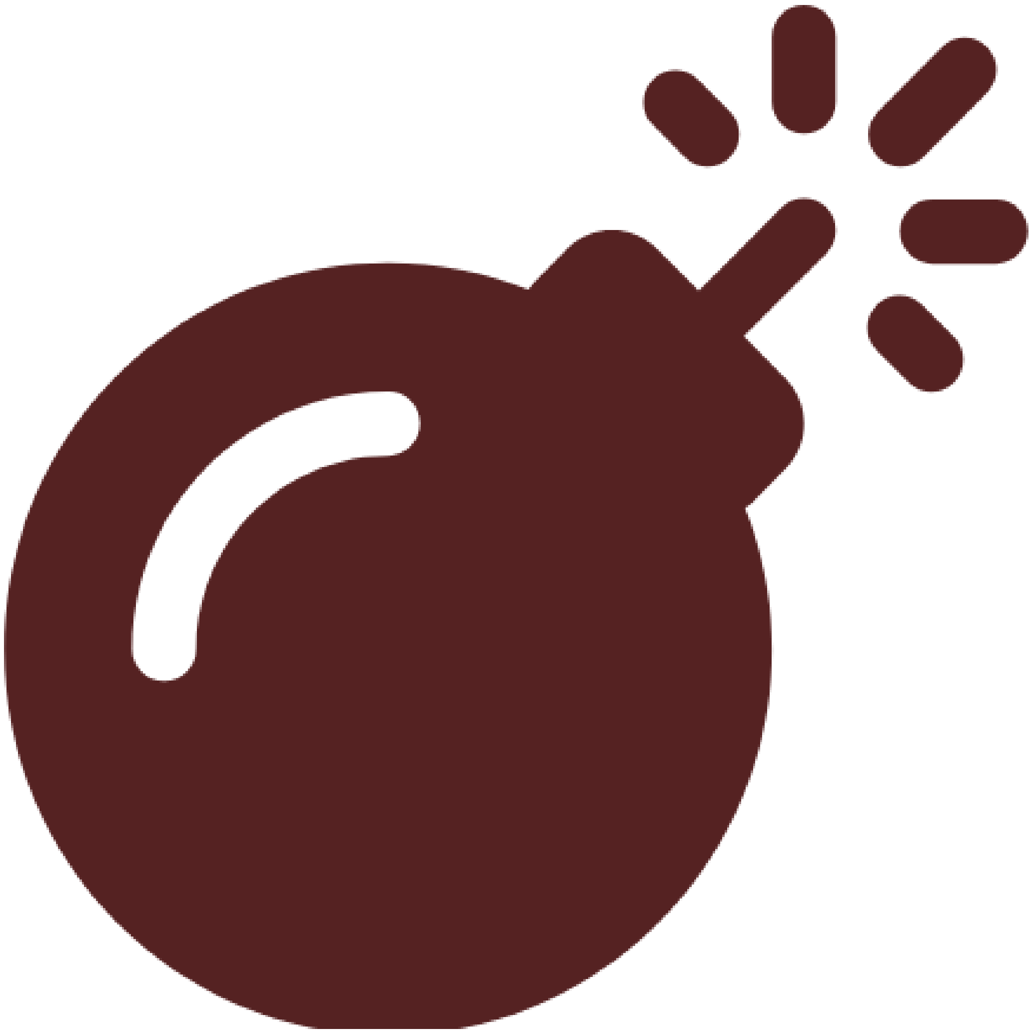
```
 _k = average of all points assigned to cluster k
```

> **Note**
>
> **Why does it converge?** Each step can only improve (or leave unchanged) the criterion. Since there's a finite number of possible assignments, the algorithm must terminate.

4. The Initialization Problem: Where to Place the First Chairs?

The Dilemma

If all chairs start in the same corner, the game will be very long and yield a poor result. **K-means is very sensitive to initialization!**



K-means++: The Smart Solution

Principle: Spread out the initial centers from each other.

Algorithm: 1. Choose a first center at random 2. For each point, calculate its distance `D(x)` to the nearest center 3. Choose the next center with probability proportional to $D(x)^2$ 4. Repeat until K centers are chosen

Why does it work? By favoring distant points, we better cover the data space.

> **Tip**
>
> **Practical tip:** Always use k-means++ or run k-means multiple times with different random initializations and keep the best result.

5. Choosing K: How Many Chairs?

The Main Problem

K-means needs to be told how many clusters to look for. But in real life, **we don't know in advance!**

Methods for Choosing K

1. Elbow Method

Principle: Plot the error (inertia) as a function of K.

```
Error(K) = sum of squared distances from points to their center
```

As K increases, the error decreases. We look for the "elbow" where improvement slows down.

2. Silhouette Score

Principle: Measures how well a point fits in its cluster vs. outside its cluster.

```
s(i) = (b(i) - a(i)) / max(a(i), b(i))
```

where `a(i)` = average distance to other points in the same cluster, `b(i)` = average distance to points in the nearest other cluster.

Interpretation: `s(i)`   `1` = well classified,   `0` = on the border,   `-1` = misclassified.

3. Practical Rule

For N points: `K`   `√(N/2)` (very rough, use with caution).

6. K-means vs GMM: The Clustering Duel

Aspect	K-means	GMM with EM
Philosophy	Geometric (distance)	Probabilistic (density)
Membership	Hard (0 or 1)	Soft (probability)
Cluster shape	Spheres (all same size)	Ellipses (variable sizes/orientations)
Model	Minimizes distance	Maximizes likelihood
Speed	Fast (O(NKT))	Slower (more calculations)
Limit case	-	K-means = GMM with $\to 0$ and spherical covariances

> **!** Important
>
> **K-means is a special case of GMM!** When we take a Gaussian mixture with: - Spherical and identical covariances: $\Sigma\_k = {}^{2}I$ - And let $\to 0$ Then soft assignment becomes hard, and we recover k-means.

7. K-means Pitfalls and How to Avoid Them

Pitfall 1: Non-Spherical Clusters

Problem: K-means assumes spherical clusters. If they're elongated or oddly shaped, it fails.

Solution: - Standardize the data (center and scale) - Use another method (GMM, spectral clustering) - Transform the data (PCA first)

Pitfall 2: Very Different Cluster Sizes

Problem: K-means tends to create clusters of similar size, even if reality is different.

Solution: Adjust K, or use a method that handles this (GMM).

Pitfall 3: Sensitivity to Outliers

Problem: A very distant point pulls its cluster center toward it.

Solution: - Detect and remove outliers beforehand - Use k-medoids (more robust) - Standardize to reduce influence

Pitfall 4: Result Interpretation

Problem: K-means always finds K clusters, even if the data has no structure.

Solution: Always validate with objective methods (silhouette, gap statistic).

8. In Practice: The Complete Recipe

Step 1: Data Preparation

1. **Clean**: handle missing values
2. **Standardize**: center and scale each variable
3. **Visualize** (if possible in 2D/3D) to get an idea

Step 2: Choosing K

1. **Elbow method**: plot inertia vs K
2. **Silhouette score**: average over all points
3. **Domain knowledge**: sometimes we know how many clusters to look for

Step 3: Execution

```
# Pseudocode
1. Initialize K centers with k-means++
2. Repeat:
   a. Assign each point to the nearest center
   b. Recalculate centers as averages
   c. If centers move little, stop
3. Evaluate with silhouette score
4. If in doubt, rerun with different initializations
```

Step 4: Validation

1. **Internal**: scores (silhouette, Davies-Bouldin)
2. **External**: if you have some labels, compare
3. **Manual**: visualization, common sense

9. K-means Variants

K-medoids

Difference: Centers are actual data points (not averages).

Advantage: More robust to outliers, works with any distance.

Mini-batch K-means

Difference: Uses data subsamples at each iteration.

Advantage: Much faster for large datasets.

Fuzzy C-means

Difference: Soft assignment (like GMM) but with a geometric criterion.

Advantage: Border points belong partially to multiple clusters.

10. Quick Review Cheat Sheet

> **!** Important
>
> **In 30 seconds for the oral exam:**
>
> 1. **K-means** = clustering by distance minimization
> 2. **Two steps**: assignment (points → nearest center) and update (recalculate centers)
> 3. **Problems**: choice of K, initialization, spherical clusters
> 4. **Solutions**: k-means++, elbow method, standardization
> 5. **Relationship**: special case of GMM with $\to 0$
> 6. **Applications**: segmentation, grouping, dimensionality reduction

11. For the Oral Exam: The Story to Tell

"Imagine organizing a party with guests scattered in a park. You want to create K discussion groups. You place K facilitators randomly (initialization), each guest joins the nearest facilitator (assignment), then the facilitators move to the center of their group (update). You repeat until the facilitators no longer move. This is exactly k-means!"

12. Frequently Asked Exam Questions

Q: “Why is k-means sensitive to initialization?” A: “Because the criterion to minimize has multiple local minima. Depending on where you start, you can get stuck in a bad minimum.”

Q: “How to choose K if you have no idea?” A: “Use the elbow method to see where improvement slows down, and validate with the silhouette score. Test several K values and see what makes sense for your data.”

Q: “When not to use k-means?” A: “When your clusters aren’t spherical, or have very different sizes, or when you have a lot of noise/outliers.”

13. Final Word

K-means is like a Swiss Army knife for clustering: **simple, fast, useful in many situations**. But like any simple tool, you need to know its limitations. The key is to always: 1. **Prepare** your data (clean, standardize) 2. **Choose K intelligently** (elbow method, silhouette) 3. **Initialize properly** (k-means++) 4. **Validate** the results (visualization, scores)

In summary: K-means is your first tool for exploring the structure of unlabeled data. Not the most sophisticated, but often the most practical to start with.